

Biology

MCQs:

1. A
2. B
3. C
4. B
5. C
6. B
7. B
8. B
9. ~~B~~ A
10. B
11. B
12. A
13. C
14. A
15. B
16. B
17. C
18. B
19. B
20. C
21. B
22. B
23. B
24. A

$$\log_2 8 = \text{?}$$

$$2 = 8$$

$$2 = (2)^3$$

$$= 3$$

Q:2(i)

A:2(i)
category:-

(i) Taxonomic

(i) Domain

(ii) Kingdom

(iii) Phylum

{ division for plants or
fungal }

(iv) Class

(v) Order

(vi) Family

(vii) Genus

(viii) species.

(b) Species :- is a group of similar organisms ~~to~~ individual capable of interbreeding or exchanging genes among themselves and producing fertile offspring.

Q:2(ii)

A:2(ii)

Structure:- DNA is a polynucleotide as it is made up of many deoxyribonucleotides and is made of 2 long stranded twisted into double helix. Each strand contains nucleotides made of sugar, phosphate group and nitrogenous base.

Complementary base pairing :-
Adenine binds with Thymine and
Cytosine pairs with Guanine.

* functions :-

DNA stores and carries genetic information that controls the living characteristics of living organisms.

Q: 2 (iii)

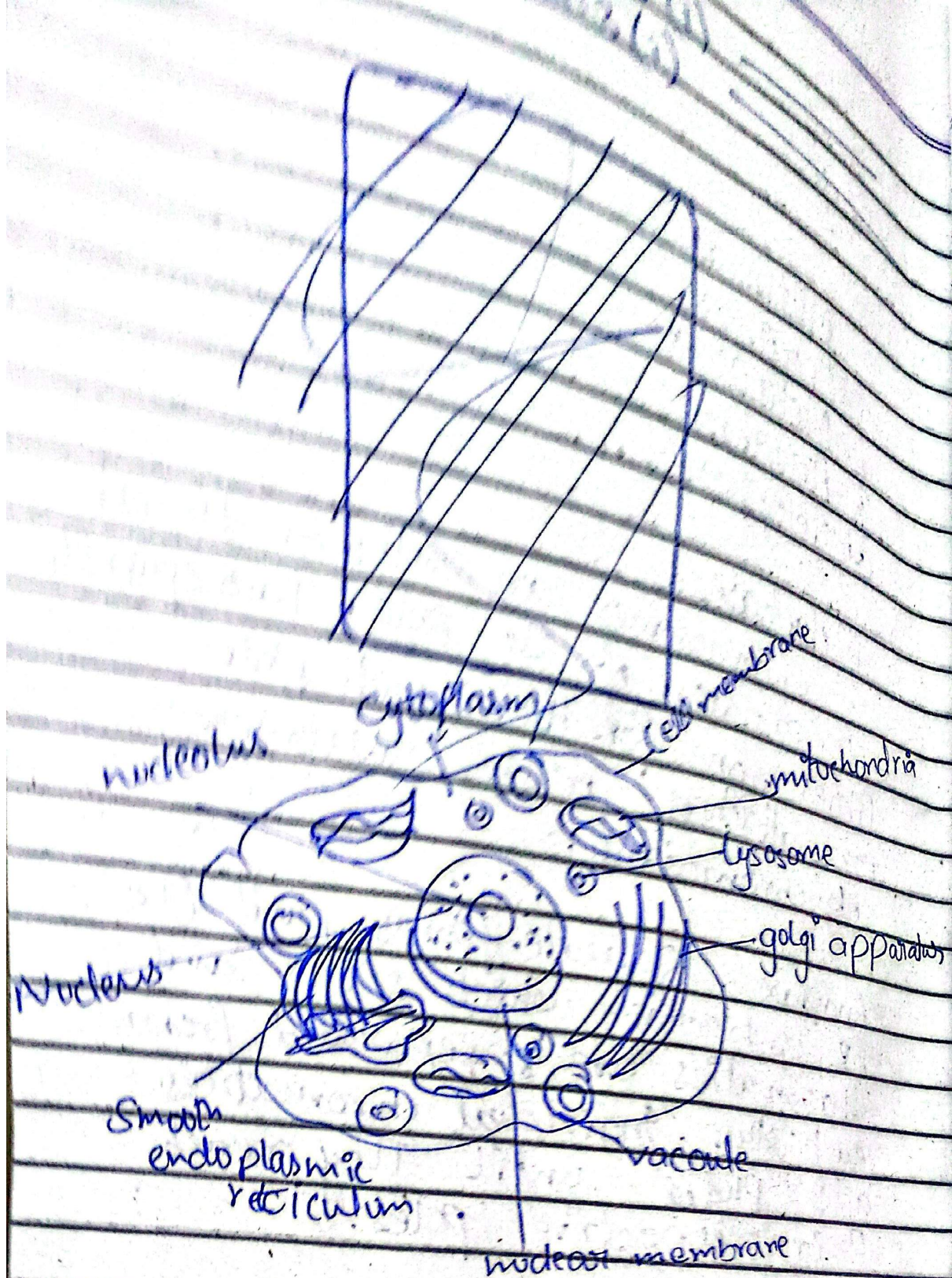
A: 2 (iii)

Three factors affecting enzyme activity :-

- * Temperature
- * pH
- * Substrate concentration

Effect of temperature :-

As temperature increases, enzyme activity increases up to optimum temperature. Above this temperature, the enzyme ~~active~~ may be denatured, changing the shape of its active site and reducing its activity.



⑤

Plant cell	Animal cell
① Chloroplast is present in plant cell	There is absence of chloroplast.
② There is cell wall	There is no cell wall.

Q: 3 (ii)
A: 3 (ii)

(a)

Mitosis :-

- ① Prophase
- ② Metaphase
- ③ Anaphase
- ④ Telophase

Metaphase :-

Chromosomes are aligned at the equator of the pole. Two spindle fibres, one from each pole are attached at the centromere of chromosome.

Anaphase :- Spindle fibres pull the centromere which splits and chromatids are separated from each other. Individual chromatids are pulled until they reach their respective poles.

Meiosis	Mitosis
① It occurs in germ cells.	It occurs in somatic cells.
② Cell divides only once	Cell divides twice.